

ASSIGNMENT NO. 3

PROBLEM STATEMENT:

Write a program in C to generate multiplication tables for a given number.

OBJECTIVES:

- To understand the use of looping statements in C.
- To generate multiplication tables using loops.
- To improve program logic and execution flow understanding.

THEORY:

Syntax for for loop example program

Syntax for while loop example program

Syntax for do while loop example program

CODE:

```
#include <stdio.h>

int main() {
    int num, i;

    printf("Enter a number: ");
    scanf("%d", &num);

    printf("\nMultiplication table for %d:\n", num);
    printf("-----\n");

    for (i = 1; i <= 10; i++) {
        printf("%d x %d = %d\n", num, i, num * i);
    }

    return 0;
}
```

OUTPUT:

```
Enter a number: 25
Multiplication table for 25:
-----
25 x 1 = 25
25 x 2 = 50
25 x 3 = 75
25 x 4 = 100
25 x 5 = 125
25 x 6 = 150
25 x 7 = 175
25 x 8 = 200
25 x 9 = 225
25 x 10 = 250
```

CONCLUSION:

Thus, the menu driven program in C successfully performs basic arithmetic operations using switch-case statements.